Detrending time series

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Computer Fundamentals for Social Sciences

This document illustrates a widely used technique in time series analysis for detrending a series (y_t). In statistics, and also in econometrics, this technique is known as the linear regression model or ordinary least squares model (OLS). This OLS technique will not be explained in detail in the following pages, primarily because the goal is to minimize formalization. Consequently, this document will limit itself to providing the essentials for a novice programmer to filter a time series based on its trend in just a few steps.

1 General time series decomposition

In general, any time series (y_t) can be written as the sum of three components, according to the equation

$$y_t = T_t + S_t + X_t. \tag{1}$$

where T_t, S_t, and X_t are commonly known as *trend*, *seasonality*, and *auto-correlation*, respectively. In particular:

- The deterministic trend (T_t) is a function of time t that displays a line toward which the time series is attracted, appearing to dance around it. This line is usually monotonic (a straight line, a quadratic line, or a cubic line) over time, but can also have less ‘regular’ trends. For example, it is known that some time series, such as a nation’s Gross Domestic Product (GDP), are expected to grow year after year. Often, this growth is evidenced by a time series that appears to follow an increasing line. In other words, in each time period, the time series appears to be attracted to this line, alternating values above and below the line while remaining close to it. In this simple example, the deterministic time trend is identified by the line;

- seasonality or cyclicity (S_t) is a periodic effect visible as an increase or decrease from the overall trend. Visually, the time series appears to oscillate smoothly around the trend, moving like waves. An example might be a country's monthly industrial production, which peaks in January and July each year, while August and December are periods of natural decline due to holiday closures;

- Autocorrelation (X_t) can be considered a 'residual' component independent of trend or seasonality. It generally consists of a sequence of numbers that can be explained by previous numbers (past values). Over time, this component follows a horizontal line above and below which the values tend to vary in each period. Furthermore, almost all the values in the sequence lie within a sort of rigid, horizontal tube. Sometimes the tube may tend to widen/shrink or exhibit a certain 'flexibility' meaning it may widen and narrow over time.

2 Detrending a time series with a linear trend

For simplicity, in this section we assume that the time series does not exhibit cyclicity ($S_t = 0$) and that the trend component is linear of the type

$$T_t = \alpha + \beta t \tag{2}$$

where $t = 1, 2, \ldots, T$, α is an intercept and β is a coefficient that can be positive (increasing trend) or negative (decreasing trend).

2.1 Some preliminary results

To detrending time series, the following well-known mathematical results are useful:

1. the sum of the integer numbers from 1 to a given value T is always given by
$$\sum_{t=1}^{T} t = 1 + 2 + 3 + \ldots + T = \frac{T(T+1)}{2}; \tag{3}$$
Hence, the sample mean of the trend is
$$\mathrm{m}_T = \frac{1}{T}\sum_{t=1}^{T} t = \frac{1}{\cancel{T}}\frac{\cancel{T}(T+1)}{2} = \frac{T+1}{2}; \tag{4}$$

2. the sum of the square integer numbers from 1 to a given value T^2 is always given by

$$\sum_{t=1}^{T} t^2 = 1 + 2^2 + 3^2 + \ldots + T^2 = \frac{T(T+1)(2T+1)}{6}. \tag{5}$$

From equations (3) and (5), it follow that the sample variance of the trend $t = [1\ 2\ \ldots\ T]'$ is

$$\begin{aligned}
\mathrm{v}_T &= \frac{1}{T-1}\sum_{t=1}^{T}(t - \mathrm{m}_T)^2 \\
&= \frac{1}{T-1}\sum_{t=1}^{T}(t^2 - 2t\mathrm{m}_T + \mathrm{m}_T^2) \\
&= \frac{1}{T-1}\left[\sum_{t=1}^{T} t^2 - 2\mathrm{m}_T \sum_{t=1}^{T} t + \sum_{t=1}^{T} \mathrm{m}_T^2\right] \\
&= \frac{1}{T-1}\left[\sum_{t=1}^{T} t^2 - 2T\mathrm{m}_T^2 + T\mathrm{m}_T^2\right] \\
&= \frac{1}{T-1}\left[\sum_{t=1}^{T} t^2 - T\mathrm{m}_T^2\right].
\end{aligned}$$

Substituting equation (4) and (5), we get

$$\begin{aligned}
\mathrm{v}_T &= \frac{1}{T-1}\left[\frac{T(T+1)(2T+1)}{6} - T\left(\frac{T+1}{2}\right)^2\right] \\
&= \frac{1}{T-1}\left[\frac{T(T+1)(2T+1)}{6} - \frac{T(T+1)^2}{4}\right] \\
&= \frac{T(T+1)}{T-1}\left[\frac{(2T+1)}{6} - \frac{T+1}{4}\right] \\
&= \frac{T(T+1)}{T-1}\left[\frac{2(2T+1) - 3(T+1)}{12}\right] \\
&= \frac{T(T+1)}{\cancel{T-1}}\frac{\cancel{T-1}}{12} \\
&= \frac{T(T+1)}{12}.
\end{aligned} \tag{6}$$

3. the sample covariance[1] between the time trend t and a time series y_t is

$$\mathrm{c}_T = \frac{1}{T-1}\sum_{t=1}^{T}(t-\mathrm{m}_T)(y_t-\bar{y}), \tag{7}$$

where $\bar{y} = \frac{1}{T}\sum_{t=1}^{T} y_t$ is the time series sample mean. After some algebra, the above covariance can be written as

$$\begin{aligned}
\mathrm{C}_T &= \frac{1}{T-1}\sum_{t=1}^{T}(ty_t - \mathrm{m}_T y_t - \bar{y}t + \mathrm{m}_T\bar{y}) \\
&= \frac{1}{T-1}\left[\sum_{t=1}^{T} ty_t - \sum_{t=1}^{T}\mathrm{m}_T y_t - \sum_{t=1}^{T}\bar{y}t + \sum_{t=1}^{T}\mathrm{m}_T\bar{y}\right] \\
&= \frac{1}{T-1}\left[\sum_{t=1}^{T} ty_t - \mathrm{m}_T\sum_{t=1}^{T} y_t - \bar{y}\sum_{t=1}^{T} t + T\mathrm{m}_T\bar{y}\right] \\
&= \frac{1}{T-1}\left[\sum_{t=1}^{T} ty_t - T\mathrm{m}_T\bar{y} - \cancel{T\mathrm{m}_T\bar{y}} + \cancel{T\mathrm{m}_T\bar{y}}\right] \\
&= \frac{1}{T-1}\left[\sum_{t=1}^{T} ty_t - \frac{T(T+1)}{2}\bar{y}\right].
\end{aligned} \tag{8}$$

2.2 Estimating the linear trend

Since in equation (2) the trend is defined by a linear function, the aim of this section is to estimate the two parameters α and β. In doing so, consider the difference

$$X_t = y_t - T_t = y_t - (\alpha + \beta t)$$

which represent the correlation component. In this setting, for each period t, the time series X_t is the deviation from the trend. From this perspective, the intercept (α) and the difference (β) must be the pair of values that make this deviation as small as possible for each t. Considering that, if the trend fits the data, the sequence X_t develops around it, it should be composed of positive and negative numbers. Consequently, the sum of all its values will

[1] In short, covariance is a statistical measure that indicates how much two variables change together. A positive covariance means the variables tend to increase or decrease in the same direction, while a negative covariance indicates they tend to move in opposite directions. A zero covariance inidcates that there is no linear relationship between the variables.

give rise to some compensation, thus minimizing the sum, which could be close to zero In practice, it can happen that

$$\sum_{t=1}^{T} X_t \approx 0$$

regardless of the parameters α and β.

To avoid this inconvenience, the solution consists in minimizing the sum of the squares with respect to the two parameters, *i.e.*

$$\min_{\alpha,\beta} \qquad \sum_{t=1}^{T} X_t^2 = \sum_{t=1}^{T} (y_t - \alpha - \beta t)^2.$$

The solutions are obtained through the first order conditions

$$\begin{cases} \dfrac{\partial \displaystyle\sum_{t=1}^{T} X_t^2}{\partial \alpha} = \displaystyle\sum_{t=1}^{T} \dfrac{\partial X_t^2}{\partial \alpha} = 0 \\ \dfrac{\partial \displaystyle\sum_{t=1}^{T} X_t^2}{\partial \beta} = \displaystyle\sum_{t=1}^{T} \dfrac{\partial X_t^2}{\partial \beta} = 0. \end{cases}$$

By calculating the partial derivatives we obtain

$$\begin{cases} -2\displaystyle\sum_{t=1}^{T} (y_t - \alpha - \beta t) = 0 \\ -2\displaystyle\sum_{t=1}^{T} (y_t - \alpha - \beta t)t = -2\displaystyle\sum_{t=1}^{T} (y_t t - \alpha t - \beta t^2) = 0 \end{cases}$$

that return

$$\begin{cases} \displaystyle\sum_{t=1}^{T} \alpha = \sum_{t=1}^{T} y_t - \beta \sum_{t=1}^{T} t \\ \displaystyle\sum_{t=1}^{T} y_t t - \sum_{t=1}^{T} \alpha t - \sum_{t=1}^{T} \beta t^2 = \sum_{t=1}^{T} y_t t - \alpha \sum_{t=1}^{T} t - \beta \sum_{t=1}^{T} t^2 = 0. \end{cases}$$

Using equations (3) and (5), we get

$$\begin{cases} T\alpha = \displaystyle\sum_{t=1}^{T} y_t - \beta \frac{T(T+1)}{2} \\ \displaystyle\sum_{t=1}^{T} y_t t - \alpha \frac{T(T+1)}{2} - \beta \sum_{t=1}^{T} t^2 = 0, \end{cases}$$

threfore, from the former equation it follows that

$$\hat{\alpha} = \frac{1}{T}\sum_{t=1}^{T} y_t - \beta\frac{\cancel{T}(T+1)}{2\cancel{T}} = \bar{y} - \beta \mathrm{m}_T, \tag{9}$$

where the apex " ˆ " indicates the estimated intercept. Substituting this result into the latter equation, we get the estimated slope

$$\begin{aligned}
&\sum_{t=1}^{T} y_t t - (\bar{y} - \beta \mathrm{m}_T)\frac{T(T+1)}{2} - \beta\sum_{t=1}^{T} t^2 = 0 \\
&\sum_{t=1}^{T} y_t t - T\bar{y}\mathrm{m}_T + \beta T \mathrm{m}_T^2 - \beta\sum_{t=1}^{T} t^2 = 0 \\
&\beta\left(\sum_{t=1}^{T} t^2 - T\mathrm{m}_T^2\right) = \sum_{t=1}^{T} y_t t - T\bar{y}\mathrm{m}_T.
\end{aligned}$$

Exploiting the variance and covariance definitions provided by equations (6) and (8), the estimated slope is

$$\hat{\beta} = \frac{\sum_{t=1}^{T} y_t t - \frac{T(T+1)}{2}\bar{y}}{\frac{T(T+1)}{2}} = \frac{\mathrm{c}_T}{\mathrm{v}_T}.$$

The estimated trend time series is

$$\hat{T}_t = \hat{\alpha} + \hat{\beta} t = \bar{y} - \frac{\mathrm{c}_T}{\mathrm{v}_T}\frac{T+1}{2} + \frac{\mathrm{c}_T}{\mathrm{v}_T} t \tag{10}$$

and the detrended time series is

$$\hat{X}_t = y_t - \hat{T}_t = y_t - \bar{y} + \frac{\mathrm{c}_T}{\mathrm{v}_T}\frac{T+1}{2} - \frac{\mathrm{c}_T}{\mathrm{v}_T} t. \tag{11}$$